

# Osaid Khan

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Dear Hiring Team,

It is with great pleasure that I submit my candidacy for the **Software Developer** with **AspenTech**. I bring over three years of professional experience in full-stack web applications, distributed systems, and AI-driven automation, along with a strong academic foundation as a Master of Science in Computer Science graduate from Pace University, NY.

Through my experience across multiple engineering roles, I have built and delivered production-grade backend systems that directly reflect **the technical depth this position demands in API architecture, massive scale data systems, and performance optimization**. As a Software Engineer at Sperse, I architected an MCP interaction layer using LangChain and LangGraph that translates natural-language intents into validated API calls across 1,000+ platform endpoints — functioning as a high-throughput data access tool for massive scale operations. To ensure the reliability of these distributed services, I established a robust observability layer using Langfuse and LangSmith, which enabled me to identify software improvements and reduce end-to-end latency by 60%. Prior to Sperse, as a Software Engineer at Qualitest, I contributed to the backbone infrastructure that thousands of users rely on daily. I designed and developed a cloud-agnostic Storage Service using Python, PostgreSQL, and Redis that handled high-volume file operations across AWS S3 and Google Cloud Storage via expiring signed URLs — providing me with direct experience in architecting APIs for complex industrial needs. I also contributed to an Identity & Access Management platform built with Java, Spring Boot, and AWS, and developed backend workflows using AWS Step Functions to process long-running data warehouse import jobs while adhering to Agile principles. Together, these roles have prepared me to **contribute meaningfully to the team's core engineering responsibilities in developing robust data access tools and maintaining high-performance historian systems** and drive impact within an agile engineering team.

My academic research experience has given me a rigorous foundation in model development, GPU-accelerated computing, and technical communication — skills that translate directly into **the engineering discipline and collaborative culture this team values at AspenTech**. As an AI Researcher at the Pace Artificial Intelligence Lab, I managed the full software development lifecycle for generative models on an HPC cluster, leveraging CUDA and PyTorch for model training and fine-tuning. I architected model inference services using vLLM, implementing daemon processes to handle high-concurrency requests and ensuring secure, efficient visualization and analysis of complex data. Beyond building the systems myself, I led workshops on GPU programming and building AI Agents for 50+ students. Designing and delivering those sessions required me to break down complex architectural stacks clearly and accessibly — directly mirroring **the technical communication and educational leadership responsibilities central to this role**. This experience has shaped me into an engineer who builds with rigor and communicates with clarity, both of which I would bring to **AspenTech and its mission of digitally transforming industries**.

I have always been passionate about **optimizing assets to run faster, longer, safer, and greener** and envision myself contributing my talent to the furthering of **autonomous, sustainable industrial operations**. I would love to discuss my interest in this position with your team and can be reached at +1 (945) 304-5781 or [khanosaid726@gmail.com](mailto:khanosaid726@gmail.com). Thank you for your time and consideration, I look forward to connecting!

Sincerely,  
Osaid Khan